

Ans 1:

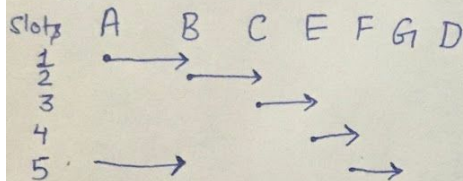
Rate control

- 1) A node should only transmit when its successor is free from interference
- 2) A node's sending rate can not exceed the sending rate of its successor

i)

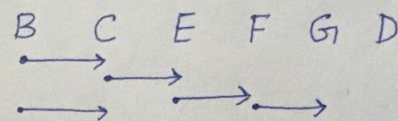
Ans 1 $\frac{2}{A}$ $\frac{3}{B}$ $\frac{1}{C}$ $\frac{4}{E}$ $\frac{1}{F}$ $\frac{1}{G}$ $\frac{1}{D}$

For A



ET of A = $R/4$

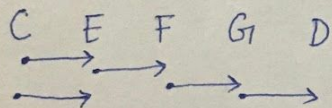
For B



B can send at $R/3$ but the source(A) itself is sending at $R/4$ so B can't send at $R/3$.

ET of B = $R/4$

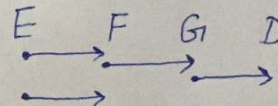
For C



ET of C = $R/4$

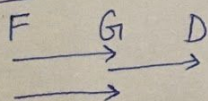
As A is creating Bottleneck.

For E



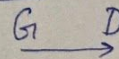
ET of E = $R/4$ (same Reason)

For F



ET of F = $R/4$

For G



ET of G = $R/4$

B, C, E, F and G can send at $R/3$, $R/3$, $R/3$, $R/2$, R respectively. They won't be able to send at this rate as A (source) itself is sending at $R/4$. So effective throughput of B, C, E, F and G will be $R/4$.

ii) source itself is sending at $R/2$ that is why even though G can send at a higher rate it won't be able to.

ii Interfering nodes for different nodes are

A - B, C

B - A, C, E, F

C → B, E

E → A, B, C, F, G, D

F → E, G

G → F, D

D - G

For A

	Slot 1	Slot 2	
ch1	A → B, F → G	B → C, G → D	ch2
ch3	C → E	E → F	ch1

ET of A = $R/2$

For B

	Slot 1	Slot 2	
ch1	B → C, G → D	C → E	ch2
ch3	E → F	F → G	ch1

ET of B = $R/2$

For F

	Slot 1	Slot 2	
ch1	F → G	G → D	ch1

ET of F = $R/2$

For C

	Slot 1		
ch1	C → E	E → F	ch2
ch2	F → G	G → D	ch1

ET of C = $R/2$

For G

	Slot 1	
ch1	G → D	

Effective Throughput = $R/2$

For E

	Slot 1	Slot 2	
ch1	E → F	F → G	ch1
ch2	G → D		

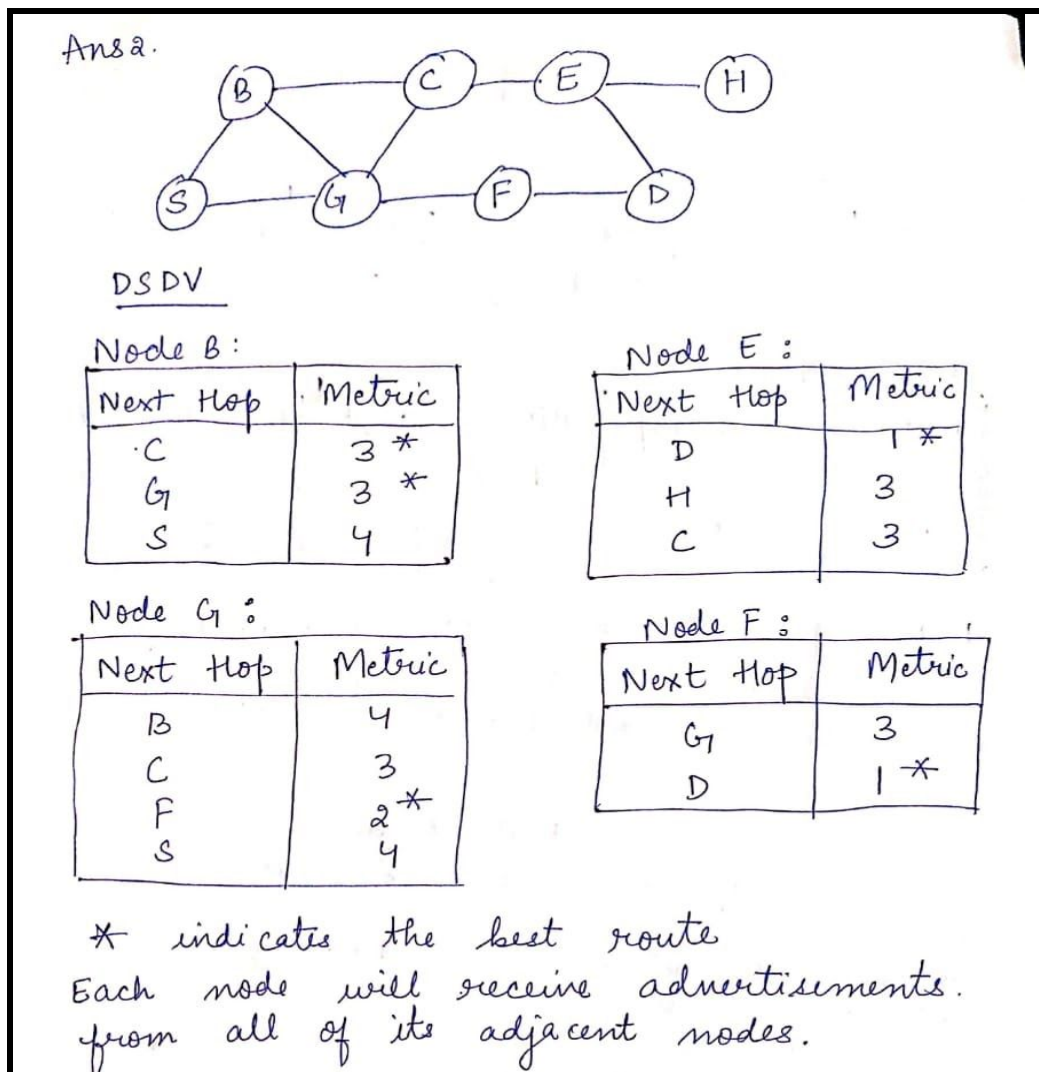
ET of E = $R/2$

G can send at ~~R~~ R but A, B, C, E and F are sending at $R/2$ so G can't send at R.

Ans 3:

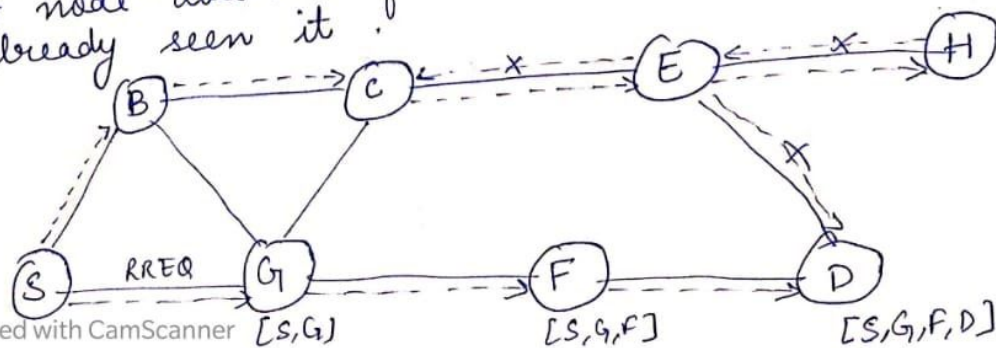
In order to avoid collision between data packets flowing from source to sink and SNACK packet from the sink to source, EOF is sent. As when EOF is sent, the source stops sending data packets and intermediate nodes clear out their buffer. Therefore, SNACK packet now faces no packets in the opposite direction.

Ans 2.

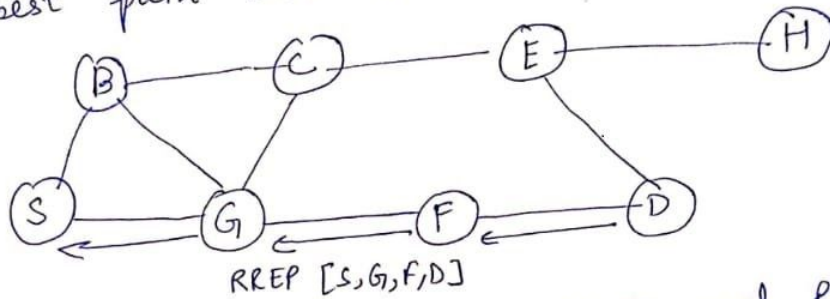


AODV - Nodes that receive the RREQ from the source set up reverse path to source.
 Nodes that receive RREP set up a forward path to the destination.

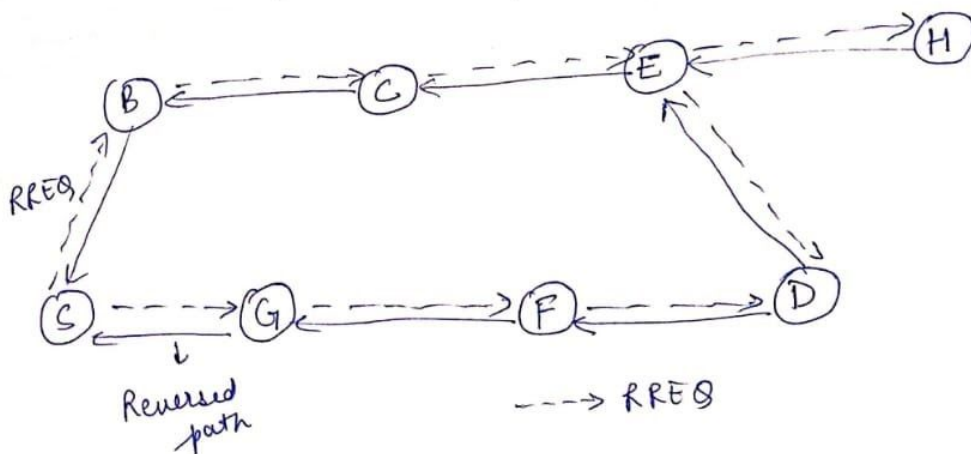
- DSR - path followed by RREQ and RREP packets.
- When a node S wants to send a packet to node D, but does not know a route to D, node S initiates a "route discovery".
 - Source node S floods Route Request (RREQ).
 - Each node appends its own identifier when forwarding RREQ.
 - A node does not forward a RREQ if it has already seen it.



- Destination on receiving the first RREQ ~~decide~~ sends a Route Reply (RREP)
- RREP is sent on a route obtained by reversing the route appended to receive RREQ
- When a route request reaches D, it picks the best path and sends back the RREP.



AODV: path followed by RREQ and RREP packet
 → Route Requests (RREQ) are forwarded in a manner similar to DSR.



When a node sends a RREQ, it sets up a reverse path pointing towards the source.

AODV assumes symmetric (bi-directional) links

When the intended destination receives a Route Request, it replies by sending a Route Reply (RREP)

Route Reply travels along the reverse path set-up when Route Request is forwarded.

Ans 4.

Ans4. In the question, there are 6 nodes.

$N_1 - N_2 - N_3 - N_4 - N_5 - N_6$

The reception ratio for packets sent from N_1 are 0%, 10%, 20%, 40%, 60% for nodes N_6, N_5, N_4, N_3, N_2 resp.

$N_6 \rightarrow X$

$N_5 \rightarrow G$

$N_4 \rightarrow G, H$

$N_3 \rightarrow E, F, G, H$

$N_2 \rightarrow A, B, E, F, G, H$

$N_1 \rightarrow A, B, C, D, E, F, G, H, I, J$

N_6, N_5, N_4, N_3 can hear themselves well
 N_5 will broadcast G after waiting for 5 packet time for N_6 .

N_6 actually do not send packets. It just sends its batchmap.

N_4 will broadcast H , after hearing G from N_5

N_3 will broadcast E, F , after hearing G, H from N_5, N_4

At this point N_6 receives G, H, E, F .

N_6 can't hear N_1 and N_2 .

N_2, N_3 can hear themselves, N_2 will broadcast A, B after hearing N_3 .

At the end of 1st round, N_6 receives G, H, E, F .



No higher priority nodes have received C, D, I, J.
Therefore, N_1 will start one more round broadcasting C, D, I, J.

Now, apply the reception ratio of 0, 10, 20, 40, 60

$$N_6 \rightarrow X$$

$$N_5 \rightarrow \frac{4 \times 10}{100} = 0.4 \Rightarrow X$$

$$N_4 \rightarrow \frac{4 \times 20}{100} = 0.8 \Rightarrow X$$

$$N_3 \rightarrow \frac{4 \times 40}{100} = 1.6 \Rightarrow 1 \text{ packet successful, assume it to be C}$$

$$N_2 \rightarrow \frac{4 \times 60}{100} = 2.4 \Rightarrow 2 \text{ packets successful, assume it to be C, D.}$$

In the 2nd round,

$$N_6 \rightarrow E, F, G, H$$

$$N_5 \rightarrow X, \text{ it did not receive any new packet}$$

$$N_4 \rightarrow X, ,$$

$$N_3 \rightarrow A, B, C \quad \begin{array}{l} \cdot A, B \text{ it heard from } N_2 \text{ (1st round)} \\ \cdot C \text{ it heard from } N_1 \text{ (2nd round)} \end{array}$$

$$N_2 \rightarrow D$$

Therefore at the end of 2nd round the destination has A, B, C, E, F, G, H.

In the 3rd round,

N_1 gets to know no higher priority nodes received I, J

$$N_5 \rightarrow \frac{2 \times 10}{100} \Rightarrow X$$

$$N_4 \rightarrow \frac{2 \times 20}{100} \rightarrow X$$

$$N_3 \rightarrow \frac{2 \times 40}{100} \rightarrow X$$

$$N_2 \rightarrow \frac{2 \times 60}{100} \rightarrow 1.2 \rightarrow 1 \text{ packet, assume it to be I}$$

$$N_6 \rightarrow A, B, C, E, F, G, H$$

$$N_5 \rightarrow X \text{ does not broadcast as it did not receive any new packet}$$

$$N_4 \rightarrow X \quad \text{? ?}$$

$$N_3 \rightarrow D \text{ it heard from } N_2 \text{ (2nd round)}$$

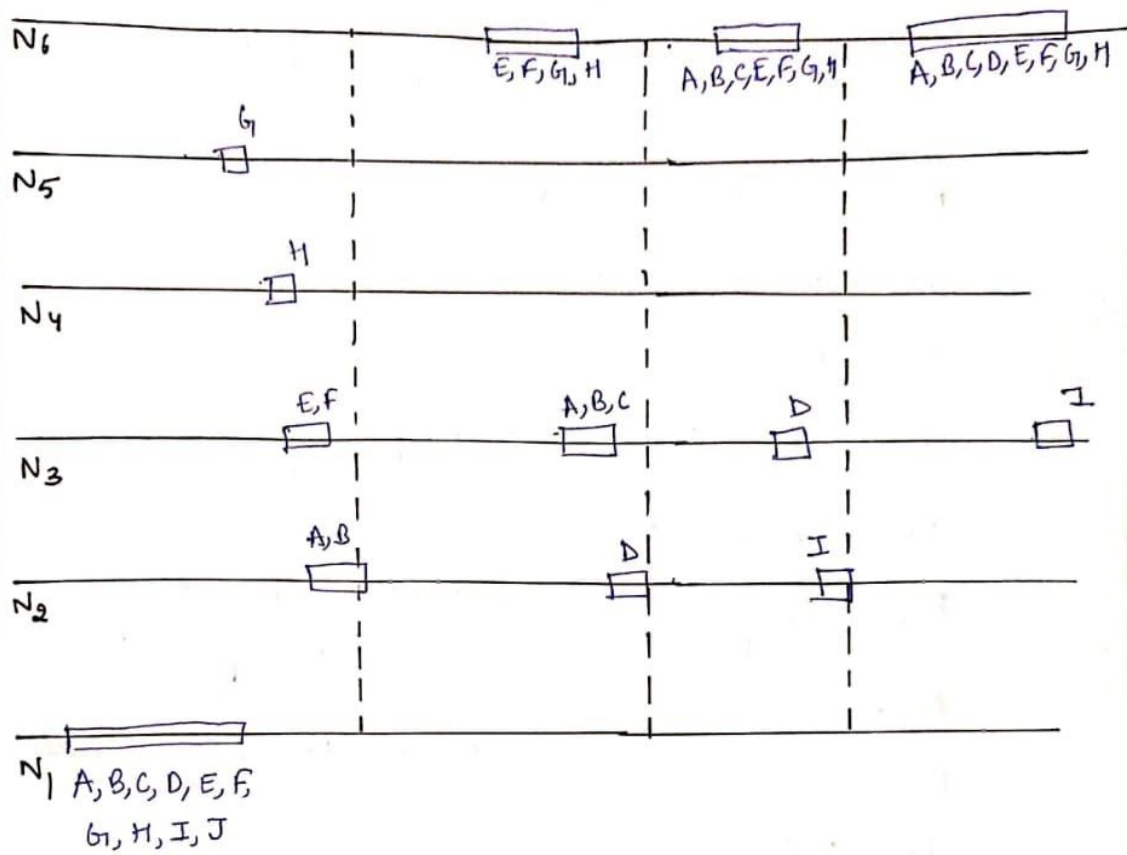
$$N_2 \rightarrow I$$

At the end of 3rd round, the destination has A, B, C, D, E, F, G, H.

Here, the source gets to know that higher priority nodes have received more than 90% of packets (A, B, C, D, E, F, G, H).

Therefore source will not send any more packets.

So, the source needs to send for 3 rounds and overall 4 rounds are needed.



Timing Diagram of ExOR transfers